

Peter Wang

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EDUCATION

University of California, Berkeley (GPA: 3.60)

May, 2026

B.S. in Computer Science & Minor in Electrical Engineering

- **Honors:** Jim & Donna Gray Endowment Award (\$10,000), Leadership Scholar (\$2,000), IEEE Member
- **Relevant Coursework:** Digital Design & Integrated Circuits, ASIC Lab, Computer Architecture, Parallel Computing, Machine Learning, Probability & Random Processes, Natural Language Processing

SKILLS

- **EDA Tools:** Synopsys DC, Cadence Genus, VCS, Innovus, Vivado, OpenROAD, gem5, Yosys, GTKWave, SKY130 PDK
- **Languages & Frameworks:** Verilog, SystemVerilog, Java, Python, C/C++, CUDA, SQL, Matlab, JavaScript
- **Libraries:** PyTorch, ONNX Runtime, Polars, NumPy, Pandas, SciPy, Matplotlib, Seaborn, scikit-learn

EXPERIENCE

Arm | SoC Architecture Intern

May 2025 — Aug. 2025

- Collaborated with physical design teams on the [Neoverse CSS](#) project to analyze pin counts and protocol mappings across IP instances for STA/timing closure.
- Built a Polars batch-inference pipeline to classify 20M+ pins and identify AMBA protocols (e.g. CHI, AXI, APB) in 15 min; cut peak RAM by 85% for SoC-scale protocol analysis.
- Mapped instance paths to die-level centroids coordinates that support across pre/post-restructuring formats.

Berkeley SLICE Lab | AWS Trainium Architecture Researcher

Aug. 2025 — May 2026

- Designing a flexible 2D systolic array for the [Zebra accelerator](#) that supports sparse matrix multiplication and dynamically adapts to dense, mildly sparse, and highly sparse workloads.
- Extending [TeAAL](#) modular architecture specifications to optimize inference kernels using Neuron Kernel Interface (NKI) with engine-aware compute mapping and manual memory scheduling.
- Conducting AWS-funded research to enable hardware-software co-design for AI workloads. *[NDA protected]*

IEEE | Lead Author & Machine Learning Engineer

Aug. 2021 — Jun. 2024

- Published research paper "[Finger-Vein Recognition using a NASNet with a Cutout](#)" in 2021 ISPACS, IEEE.
- Utilized CUDA for parallelized training, implementing custom CUDA kernels to accelerate convolutional layers and matrix multiplications, reducing model training time by 36%.

Honda Research Institute | AI/ML Intern

Jan. 2024 — May 2024

- Analyzed 100K+ telematics sessions using L2 Ridge Regression and K-means clustering to model EV energy usage and propose route adjustments for congestion relief.
- Co-authored capstone research project: "[Potential of Telematics Data in Data Science](#)" (PDF)

CHIP DESIGN

Sparse Tensor Accelerator Modeling ([Trapezoid](#)) | TeAAL, Python

- Modeled the Trapezoid accelerator in TeAAL, simulating a 2D spatial array with multi-fiber intersection units to efficiently handle varying sparsity levels (Dense vs. Highly Sparse).
- Evaluated Gustavson-based versus inner-product dataflows, optimizing fiber-tree mappings to maximize PE utilization and minimize off-chip memory traffic.

3-Stage Pipelined RISC-V CPU & Cache | Verilog, SystemVerilog

- Designed a 3-stage pipelined RISC-V CPU in Verilog with complete data forwarding, achieving **1.05 CPI**.
- Integrated a GShare branch prediction unit with a 2-bit PHT and tag-indexed BTB, improving prediction accuracy and reducing control-path stalls for better performance.
- Built a **2-way set-associative write-back cache** with valid/dirty bits and LFSR-based pseudo-random replacement policy, yielding 40%+ speedup over direct-mapped.
- Verified via **SystemVerilog** assertions and synthesized RTL using **SKY130 PDK** for timing closure and area optimization in an RTL-to-GDS workflow.

TSMC 16nm Tapeout

- Expected Spring 2026